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Experiment Name:Random Sampling and Sampling Distribution

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import numpy as np
import matplotlib.pyplot as plt

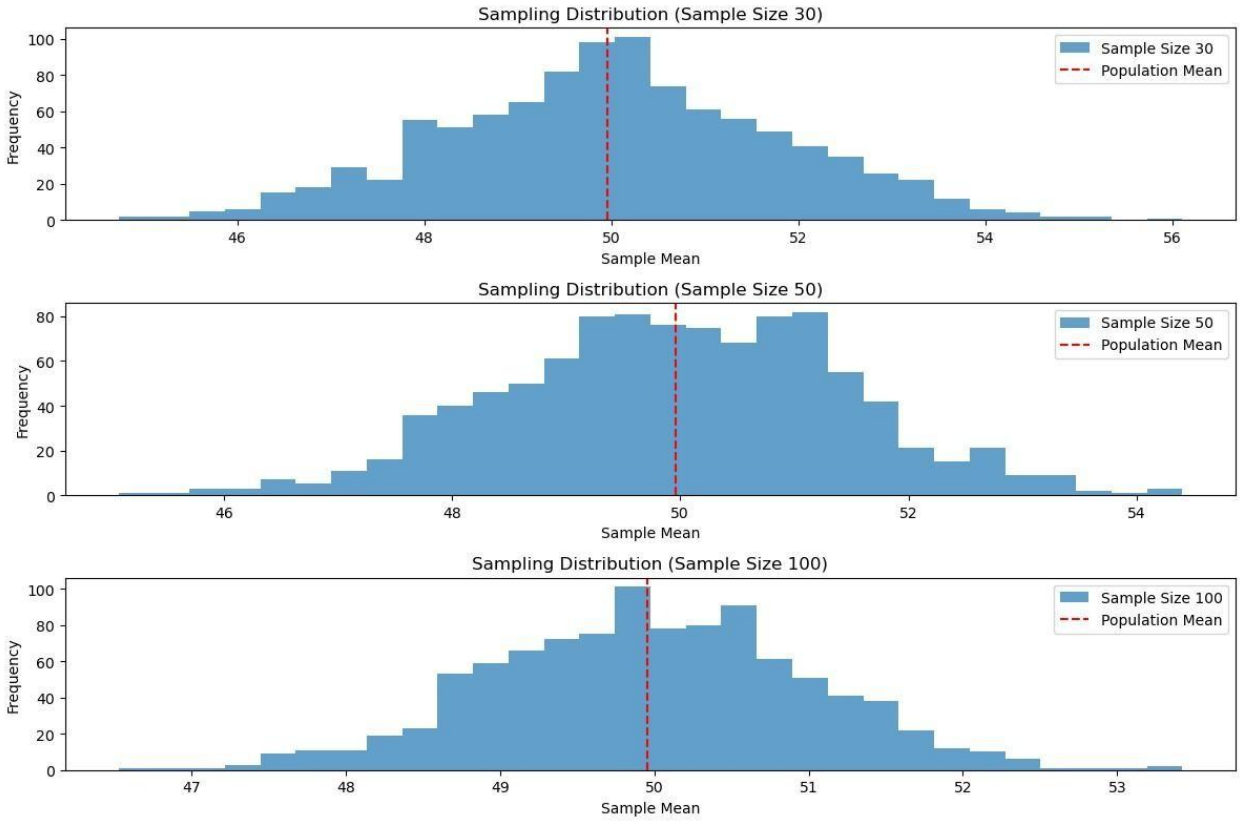
# Step 1: Generate a population (e.g., normal distribution)
population_mean = 50 population_std = 10 population_size =
100000
population = np.random.normal(population_mean, population_std,
population_size)

# Step 2: Random sampling
sample_sizes = [30, 50, 100] # different sample sizes to consider
num_samples = 1000 # number of samples for each sample size
sample_means = {}

for size in sample_sizes:     sample_means[size] = []
for _ in range(num_samples):          sample =
np.random.choice(population, size=size, replace=False)
sample_means[size].append(np.mean(sample))

# Step 3: Plotting sampling distributions
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
plt.subplot(len(sample_sizes), 1, i+1)
    plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size
{size}')
    plt.axvline(np.mean(population), color='red', linestyle='dashed',
linewidth=1.5, label='Population Mean')
    plt.title(f'Sampling Distribution (Sample Size {size})')
plt.xlabel('Sample  Mean')           plt.ylabel('Frequency')
plt.legend() plt.tight_layout() plt.show()
```


Sampling Distribution (Sample Size 30)
Frequency
Sample Mean
Sample Size 30
Population Mean
0
20
40
60
80
100
46
48
50
52
54
56

Sampling Distribution (Sample Size 50)
Frequency
Sample Mean
Sample Size 50
Population Mean
0
20
40
60
80
46
48
50
52
54

Sampling Distribution (Sample Size 100)
Frequency
Sample Mean
Sample Size 100
Population Mean
0
20
40
60
80
100
47
48
49
50
51
52
53